

Pavo Cloud

Technical Execution Report | AI for Business Track

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1. What Changed Since Round 1

1.1 From an Argument to Working Software

Round 1 argued that prior authorization is slow because a person has to start every request. This report replaces that argument with the software itself. All four planned phases are built and running, and every figure here was measured on the code on 13 September 2026 rather than estimated.

Table 1. What was built, in the order it was built.

Phase	What it does	Status
Phase 1	The core loop: a doctor's order becomes an approval with no person involved	Complete
Phase 2	Insure: shows a patient what a procedure will actually cost them	Complete
Phase 3	Identity and appeals: agents prove who they are, and denials are appealed automatically	Complete
Phase 4	Privacy proofs: show a patient qualifies without sending their medical details	Complete

1.2 What the System Does, Measured

Round 1 promised a decision in under five minutes. The software turned out to be far faster, because the slow part was never the thinking — it was the paperwork around it.

Table 2. Measured results. A millisecond is one thousandth of a second.

What was measured	Result
A complete authorization decision, start to finish	88.6 milliseconds
The two agents agreeing, without the network in between	44.7 milliseconds
Creating a privacy proof about a patient	406 milliseconds
The full six-step demonstration, front to back	11.1 seconds
Automated tests passing, out of 138	138

1.3 Two Corrections to Round 1

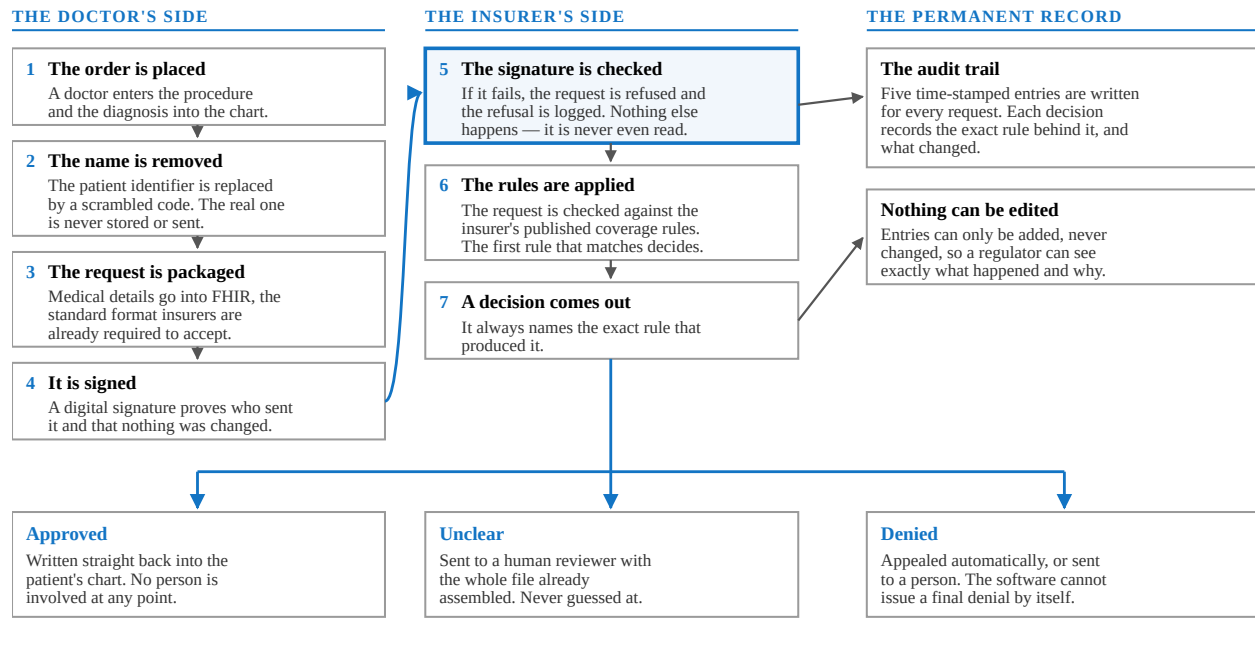
Two claims did not survive being built, and it is better to say so here than to be asked. On *speed*, Round 1 said “under five minutes”; the agents themselves take under a tenth of a second, so five minutes now covers the round trip once the hospital's record system and the insurer's network are included. On *profitability*, Round 1 implied the company breaks even on each sale in year one. Rebuilt from what the software costs to run, year one is a 31% gross margin — 31 cents of every

dollar left after the direct cost of serving that customer — because early customers need a great deal of hand-holding. The 87% figure for year three held.

2. System Architecture 2.0

2.1 How One Request Moves Through the System

A doctor places an order in the hospital's electronic health record — the software that holds a patient's chart. That order starts everything below. No person touches any of the seven steps.



Typical time from step 1 to step 7: **88.6 milliseconds**, measured over twenty consecutive runs. No person touches any of the seven steps. The blue path is the only one that crosses between two organisations.

Figure 1. One authorization request, end to end. The heavier box at step 5 is a gate: if the signature does not check out, the insurer's software never reads the request at all.

2.2 Why Step 5 Matters

Two organisations letting software talk automatically only works if each can prove who it is. Step 5 is where that happens, and it happens *before* the request is opened: either the sender is proven genuine, or nothing happens. A judge can test this by changing any part of a request after it has been signed — it is refused outright.

2.3 Two Layers, and Which One Is Allowed to Decide

Round 1 described a fixed-rules layer and a learning layer. The built system draws that line harder than promised. **No AI language model ever decides an authorization.** Models are used only where being approximately right is acceptable — reading a photograph of an insurance card, for instance — and every result is scored before anything is done with it.

Table 3. What each layer is allowed to do.

Layer	What it handles	The guarantee it gives
Fixed rules (decides)	Identity checks, coverage rules, price arithmetic, sorting denial reasons	The same input always produces the same answer, and every answer names the exact rule behind it. That is what makes a decision defensible to a regulator.
AI models (assist only)	Reading insurance cards and benefit documents, turning plain English into a medical code, drafting appeal letters	Every result carries a confidence score. Below the threshold it goes to a person. A model is never the last word on whether a patient gets care.

One consequence is worth stating plainly. If the AI provider went offline tomorrow, prior authorization would keep working, because there is no AI on that path. The same fact explains why processing one request costs a fraction of a cent (section 5.2).

2.4 When a Request Is Denied

A denial is never the end. The software sorts the insurer's stated reason, then acts on it.

Table 4. What happens after a denial.

Reason given	What the software does
Not covered by the plan	Stops. This is a contract question, not a medical one, so a person decides. No letter is drafted.
Missing information, not medically necessary, or any unrecognised reason	Searches the published medical literature, drafts an appeal citing the studies it finds, and scores its own confidence in it.
Confidence below 70%	The appeal is held for a person, with the diagnosis, the reason, the score and every study already attached.

2.5 What Happens to Patient Data

Patient identifiers are scrambled at the door, before anything is saved, and the scrambling is one-way: the original cannot be recovered from it. A test in the codebase checks that the real identifier never appears in storage.

The system goes further using a technique called a zero-knowledge proof. It lets the insurer confirm three things — that the patient is old enough, that the diagnosis is one the plan covers, and that the deductible condition is met — *without receiving the age, the diagnosis or the amount*. The insurer learns only whether each condition holds. One limitation is listed in Table 7.

3. The Build Log

3.1 What Was Committed, and When

The code lives in a repository called *hwangus922/AI2O-Pavo-Cloud*. Work was done in five batches — one per phase, plus a round of fixes — and each was reviewed before being merged. Together they come to 19,541 lines of code across 170 file changes.

```
$ git log --first-parent --date=short --pretty="%h %ad %s" a13f6cb

a13f6cb 2026-09-12  Fix the three failures that break a deployed demo (#5)
abc1b5c 2026-09-09  Phase 4: demo flow, audit UI, and a real ZK proof (#4)
99bbb96 2026-09-08  Phase 3: autonomous appeals and cryptographic identity (#3)
5f26194 2026-09-07  Phase 2: Insure price transparency (#2)
e48ec54 2026-09-07  Phase 1: autonomous prior authorization core loop (#1)
d5ea3a7 2026-09-07  Initial commit

$ git log --shortstat a13f6cb

12 commits · 170 file changes · +19,541 insertions · -367 deletions
```

Figure 2. The commit history. Each line is a batch of work that was reviewed and merged. Phase 1 was the largest at 9,429 lines; the last entry fixes three problems that only appear once the software is deployed rather than run on a laptop.

3.2 The Tests

An automated test is a small program that checks the main program still behaves correctly. There are 138 of them and they all pass. They matter more than the line count, because they are what turns a claim in this report into something a judge can verify with one command.

```
$ .venv/bin/python -m pytest

.....
..... [ 52%]
.....
..... [100%]
138 passed, 1 warning in 22.37s
```

Figure 3. The full test suite; each dot is one passing test. Among the things they prove: every decision records the rule that produced it, an unmatched request goes to a human rather than being guessed at, a tampered message is refused, a real patient identifier never reaches storage, and a rejected appeal is escalated rather than becoming an automatic final denial.

3.3 The System Running

The screens below are the live software, not mock-ups. Both were captured on 13 September 2026.

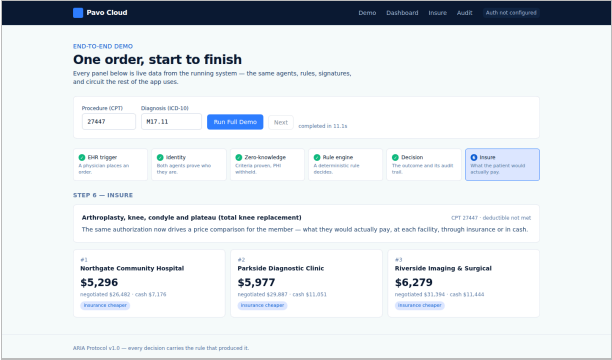


Figure 4. The guided demonstration after a single click. All six steps — order, identity, privacy proof, rules, decision, price — run with no further input and finish in about eleven seconds. The last step prices the same operation at five facilities: a spread of \$2,475 for identical care.

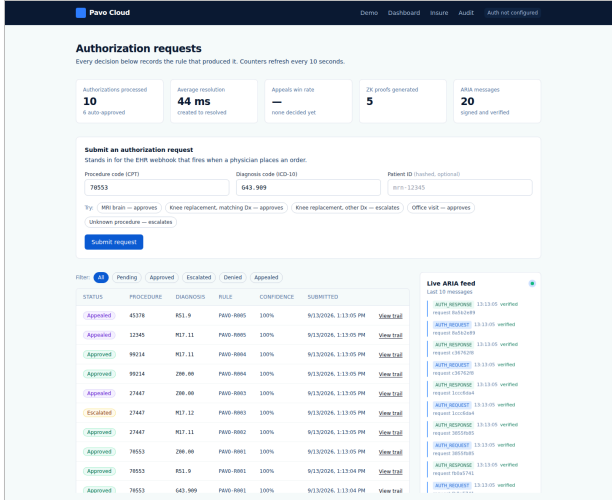


Figure 5. The dashboard. Every row carries the rule that produced it, in the RULE column. The single amber row is a knee replacement whose diagnosis did not match the covered condition; it is waiting for a human reviewer, with its file already assembled.

4. Risk Mitigation Protocol

4.1 The Governing Rule

One principle runs through every part of the system: **the safe outcome is the default, not the exception.** When the software is unsure it does not guess. It stops and hands the case to a person, with the file already prepared.

Table 6. What happens when something goes wrong.

If this happens	The software does this, automatically	A person is involved
No coverage rule matches the request	Marks it unclear and routes it to a human reviewer. There is no path by which an unmatched request is approved or denied.	Within 4 hours, insurer's clinical reviewer
A message has been tampered with	Refuses it before reading it, and records the refusal with the reason.	Immediate security alert
An appeal scores below 70% confidence	Holds it. Nothing is sent to the insurer. Reviewer notes are written automatically.	Within 1 working day, provider's appeals staff
The AI provider goes offline	Appeal drafting falls back to a placeholder scored zero, which is below the threshold, so every appeal escalates. Authorization itself is unaffected.	None required
The medical literature service is unreachable	Finishes the appeal with no citations and records the failure. Confidence drops, which pushes it to a person.	Follows the appeal route above

An appeal is rejected by the insurer	Marks it unclear. The software cannot issue a final denial to a patient.	Always, a licensed reviewer
Approval rates drift by demographic	Quarterly review across age, geography and diagnosis. Any rule causing a disparity is suspended to human review.	Quarterly, clinical and compliance

4.2 Why Human Review Is Not a Queue

Escalating a case is a routing decision, not a backlog. Every case handed to a person arrives complete: the medical details, the rule that fired, the signature check, the denial reason, the confidence score and any studies found, all assembled before the reviewer opens it. The bottleneck Round 1 identified was never medical judgement — it was the paperwork wrapped around it.

4.3 What We Have Not Solved

These are real limitations of a prototype. We would rather state them than be caught by them.

Table 7. Known gaps and what each needs.

Gap	What it needs
The privacy proof reveals which covered condition a patient has	A more advanced circuit design. Scoped for the next phase.
The proof's setup was done on one machine	A proper multi-party ceremony before any real use. Not safe for production as it stands.
Provider identity is trusted, not verified	A live lookup against the federal provider registry. One integration.
Facility prices are generated, not gathered	The voice agent that calls facilities, plus the price files insurers must now publish.
Five coverage rules, not a rule library	Clinical staff encoding each insurer's published criteria. This is the largest cost in our model and the real barrier to a competitor.
The learning layer is not built	Deliberate. Until it exists, anything without a definitive rule goes to a person — which is the correct default anyway.

5. Three-Year Financials

5.1 Where the Money Comes From

Pavo earns in three ways: a fee for each request, a monthly subscription per medical practice, and a yearly fee to insurers for the connection itself. The figures below are built up from customer counts and request volumes, not from a growth rate applied to a starting number. The full workbook accompanies this report.

Table 8. Revenue, built up from customers.

Source	2026	2027	2028
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Medical practices served (average)	55	330	1,400
Insurers connected (average)	11	84	160
Requests processed	171,600	1,647,360	8,736,000
Fees per request, at \$3.00	\$514,800	\$4,942,080	\$26,208,000
Practice subscriptions, at \$400/month	\$264,000	\$1,584,000	\$6,720,000
Insurer connection fees, at \$20,000/year	\$220,000	\$1,680,000	\$3,200,000
Total revenue	\$998,800	\$8,206,080	\$36,128,000

The \$3.00 fee is 85–92% below the \$15–\$40 an insurer spends handling a request by hand today, which is what makes the switch easy to justify.

5.2 What It Costs to Run

Servers are not the expense. Processing one request costs about six hundredths of a cent, because the deciding path uses no AI models at all. The real cost is clinical staff translating each insurer's published rules into the fixed rules the software applies — the largest line in every year, and the reason margins improve with scale rather than with technology.

Table 9. Cost of serving customers, and what is left.

Cost	2026	2027	2028
Clinical staff writing coverage rules	\$189,000	\$455,000	\$1,080,000
Setting customers up	\$224,000	\$476,000	\$1,428,000
Customer support	\$92,000	\$190,000	\$784,000
Servers and storage	\$54,000	\$198,000	\$900,000
AI model usage (documents and appeals only)	\$11,000	\$62,000	\$280,000
Security and compliance certification	\$118,000	\$112,000	\$290,000
Gross profit	\$310,800	\$6,713,080	\$31,366,000
Gross margin	31%	82%	87%

5.3 Cost to Win a Customer, and What One Is Worth

Two standard measures. *Acquisition cost* is what it costs in sales and marketing to sign one customer. *Lifetime value* is the profit that customer produces before they leave, stated here over three years, which is deliberately conservative.

Table 10. Customer economics at 2028 rates.

Measure	Practice	Insurer	Blended
Cost to acquire one customer	\$6,000	\$22,000	\$6,699
Value over three years	\$58,248	\$51,057	\$57,533
Value per \$1 spent acquiring	\$9.70	\$2.30	\$8.60

Months to earn the cost back	3.5	15.2	4.0
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Read the insurer column honestly. At \$2.30 back per \$1 spent and fifteen months to recover it, insurer contracts do not pay for themselves as a product. We fund them anyway, because signing one insurer makes Pavo available to every practice that already submits to it. It is a distribution channel, not a profit centre, and it is why the cost of winning a practice falls 37% across the plan.

5.4 Profit, and What We Are Asking For

After engineering, sales and admin, the company loses \$2.37 million in 2026 and \$3.21 million in 2027, then makes \$11.23 million in 2028. The deepest point is \$5.58 million of cumulative losses, in late 2027.

We are asking for \$8.0 million, which covers that with roughly a year of cushion beyond breaking even. In order of how much risk each removes, it buys the clinical rule library (\$1.7m), the security certification no insurer will sign without (\$520k), insurer business development (\$5.67m), and the engineering to close the gaps in Table 7 (\$7.46m).

The assumption most likely to be wrong is the number of practices, not the price. If all four of our main assumptions are wrong at once, 2028 revenue is \$13.8 million rather than \$36.1 million: a smaller company, but still a real one.